

Practical1

The image displays three screenshots of the CODETANTRA online IDE interface, showing Python programs for different tasks. Each screenshot includes a problem description, input/output formats, and a sample test cases section.

1.1.1. Calculate Momentum

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:

- m is the mass of the object (in kilograms).
- v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

```
1 m=float(input(""))
2 v=float(input(""))
3 p=m*v
4 print(f"{p:.2f}kgm/s")
```

1.1.2. Conditional Calculation Based on the Number of Digits

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:

- $1 \leq n \leq 999$

Input Format:

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

```
1 import math
2 n=int(input(""))
3 if abs(n)<10:
4     result = n**2
5     print(result)
6 elif abs(n)<100:
7     result = math.sqrt(n)
8     print(f"{result:.2f}")
9 elif abs(n)<1000:
10    result = n**(1/3)
11    print(f"{result:.2f}")
12 else:
13    print("Invalid")
```

1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format $YYYY-MM-DD$.
- The second line contains the second date in the format $YYYY-MM-DD$.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

```
1 from datetime import datetime
2
3 date1 = input().strip()
4 date2 = input().strip()
5
6 d1 = datetime.strptime(date1, "%Y-%m-%d")
7 d2 = datetime.strptime(date2, "%Y-%m-%d")
8
9 diff = (d2 - d1).days
10
11 print(diff)
```

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1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

reversen...

```
1 n = int(input())
2 reversed_n = int(str(n)[::-1])
3 print(reversed_n)
```

Terminal Test cases

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1.1.5. Multiplication Table

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:
`(number) x (multiplier) = (result)`

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

multiplica...

```
1 n = int(input())
2 for i in range(1, 11):
3     result = n*i
4     print(f"{n} x {i} = {result}")
5
6
7
```

Terminal Test cases

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1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses. The grade is determined based on the aggregate percentage.

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

- If the student passes all courses:
 - Print the aggregate percentage (formatted to two decimal places).
 - Print the grade based on the aggregate percentage.
- If the student fails any course (marks < 40 in any course), print:
 - "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Sample Test Cases

passorFa...

```
1 n = int(input())
2 marks = list(map(int, input().split()))
3
4 if len(marks) != n:
5     print("Invalid Input")
6 else:
7     if any(mark < 40 for mark in marks):
8         print("Fail")
9     else:
10        aggregate = sum(marks) / n
11        print(f"Aggregate Percentage: {aggregate:.2f}")
12        if aggregate >= 75:
13            grade = "Distinction"
14            print(f"Grade: {grade}")
15        elif 60 <= aggregate < 75:
16            grade = "First Division"
17            print(f"Grade: {grade}")
18        elif 50 <= aggregate < 60:
19            grade = "Second Division"
20            print(f"Grade: {grade}")
21        elif 40 <= aggregate < 50:
22            grade = "Third Division"
23            print(f"Grade: {grade}")
24        else:
25            print("Fail")
26
```

Terminal Test cases

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1.2.2. Fibonacci Series

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

```
1 def fibonacci(n):
2     if n == 1:
3         return 0
4     elif n == 2:
5         return 1
6     else:
7         return fibonacci(n-1) + fibonacci(n-2)
8
9 n = int(input())
10 for i in range(1, n+1):
11     print(fibonacci(i), end=" ")
12
13
```

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1.2.3. Pattern - 1

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

```
1 n = int(input())
2 for i in range(1, n+1):
3     print("*" * i)
```

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1.2.4. Pattern - 2

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

```
1 n = int(input())
2 for i in range(1, n+1):
3     for j in range(1, i+1):
4         print(j, end=" ")
5     print()
```

Terminal

Test cases